

BTC/USD Trading Strategy — Backtest Report

Trend-Following Strategy (Donchian Breakout + EMA/ADX Filter + ATR Trailing Stop)

1. Strategy Logic

My hypothesis was that BTC/USD doesn't move randomly all the time, it usually goes through long stretches of strong trending moves broken up by choppy, sideways periods. A plain breakout system would get chopped up badly during the sideways parts, so I added two filters on top of a 20-day Donchian channel breakout. First, the breakout has to agree with the trend direction given by a 20/50 EMA crossover (fast EMA above slow EMA for longs, below for shorts). Second, ADX(14) has to be above 20, which means the market actually has some directional strength behind it and isn't just drifting. Once a trade is on, I don't use a fixed target, I use a 3x ATR trailing stop that moves with the position so winning trades can run further while losing trades get cut off early. The strategy can also flip directly from long to short (and vice versa) if a strong opposite breakout shows up before the trailing stop is hit.

2. Key Performance Metrics

Backtested on daily BTC/USD data (2018–2022), \$1000 starting capital, compounded, 0.15% brokerage fee per trade. No lookahead bias was detected when re-running the strategy signal by signal on truncated data.

Metric	Value
Total Trades	23
Winning / Losing Trades	13 / 10
Win Rate	56.5%
Net Profit (on \$1000)	\$8,909.90
Benchmark Return (Buy & Hold)	23.6% (\$236.35)
Sharpe Ratio	1.15
Maximum Drawdown	37.65%
Average Drawdown	7.41%
Largest Win / Largest Loss	\$3,402.03 / -\$1,448.56
Winning Streak / Losing Streak	4 / 2
Average Holding Time	46 days

The strategy massively beat the buy-and-hold benchmark on this training window, mostly because compounding on a small number of large trending moves (the trade sizing grows with the account) amplifies returns a lot. That said, a 37.65% max drawdown is fairly large, and 23 trades over 5 years

isn't a huge sample size, so these numbers should be read with some caution rather than as a guarantee of how it performs on unseen data.

3. Parameter Selection

The two main knobs in the strategy are the ADX threshold (how strong a trend has to be before a breakout is taken) and the ATR multiplier on the trailing stop (how much room a trade is given before it's cut). I swept both and re-ran the full backtest for each combination rather than just picking numbers that felt right.

ATR Mult	ADX Th.	Trades	Sharpe	Net Profit	Max DD	Note
2.0	15	36	1.05	\$6,012	29.3%	
2.0	20	33	1.01	\$5,252	29.3%	
2.0	25	26	0.52	\$888	51.0%	ADX filter too strict
3.0	15	26	1.08	\$7,567	37.7%	
3.0	20	23	1.15	\$8,910	37.7%	Best
3.0	25	21	0.42	\$546	59.0%	ADX filter too strict
4.0	15	22	1.07	\$8,377	49.2%	
4.0	20	20	1.02	\$7,038	49.2%	
4.0	25	18	0.45	\$606	74.3%	ADX filter too strict

ADX 25 hurts performance across every ATR setting, it filters out too many trades and the strategy starts catching only the very tail end of moves. ADX 20 with a 3x ATR stop gave the best Sharpe (1.15) and the best net profit, so that's the combination used in the final version. A tighter 2x ATR stop cuts drawdown but also cuts winners short, and a wider 4x stop increases drawdown without really improving Sharpe, so 3x was the middle ground.

4. Limitations

Breaking the net profit down by year shows the strategy lost a bit in 2018 (-\$233.88) and did okay but not great in 2019-2020 (+\$731.50, +\$597.60). Almost all of the profit actually came from 2021 alone (+\$5,744.77), with 2022 adding another +\$2,069.92. So the strong Sharpe ratio is being carried heavily by one very strong trending year, and a lot less by the choppy years. This means the backtest numbers could look a lot weaker on a test period that doesn't have a 2021-style trend in it. With only 23 total trades the sample size is also small, so individual trades have a big effect on the overall stats. The strategy is also inherently a trend-follower, it will underperform in sideways/ranging markets even with the ADX filter on, since ADX only reduces bad entries, it doesn't eliminate them.

5. Comparison / Ranking

Only one strategy is being submitted here, so there is no ranking to report.